

Sustainable Animal and Aquatic Foods Science Program

Inception Report

June 2025

Contact:

Interim Deputy Director: Karen Marshall - K.Marshall@CGIAR.org

Interim Director: Rodrigue Yossa - R.Yossa@cgiar.org

Table of Contents

List of acronyms	3
Executive summary.....	4
1. Program/Accelerator overview.....	5
1.1 High-level Vision	5
1.2 Achievements from 2022–2024	5
1.3 Description of intended research	6
2. Co-design and partnerships	8
3. Theory of change and MELIA	10
3.1 Theory of Change	10
3.2 Monitoring, Evaluation, Learning, and Impact Assessment (MELIA)	12
4. Comparative advantage analysis	13
5. Prioritization.....	14
5.1 Knowledge-driven priorities	14
5.2 Final decisions on the Program/Accelerator’s direction and design	15
6. Alignment of W3 and bilaterally funded work	17
7. Plan of Results and Budget	18
8. Cross-Portfolio linkages and geographic coordination.....	20
8.1 Cross-portfolio linkages	20
8.2 Geographic coordination.....	20
9. Risk management.....	21
10. Addressing feedback on select topics	23
Appendix 1: Addressing specific feedback.....	25

List of acronyms

AAF	Animal and Aquatic Food
ABS	Access and Benefit Sharing
AMR	Antimicrobial resistance
AMU	Antimicrobial use
AoW	Area of Work
ARI	Advanced Research Institutes
ASEAN	Association of Southeast Asian Nations
BDN	Better Diets and Nutrition
CASP	Climate Action Science Program
CBO	Community-Based Organizations
CGIAR	Consultative Group on International Agricultural Research
CIAT	Alliance of Bioversity International and CIAT
CWANA	Central and West Asia and North Africa
EA	Evaluability Assessment
ESA	East and Southern Africa
GYSI	Gender, Youth, and Social Inclusion
HLO	High-Level Outcomes
ICARDA	International Center for Agricultural Research in the Dry Areas
ILRI	International Livestock Research Institute
ISDC	Independent Science for Development Council
IWMI	International Water Management Institute
L&C	Livestock and Climate
LAC	Latin America and the Caribbean
MEL	Monitoring, Evaluation and Learning
MELIA	Monitoring, Evaluation, Learning, and Impact Assessment
MFL	Multifunctional Landscapes
MRV	Measurement, Reporting, and Verification
NARES	National Agricultural Research and Extension Systems
PO	Program Outcomes
PORB	Plan of Results and Budget
PRMS	Performance and Results Management System
SA	South Asia
SAAF	Sustainable Animal and Aquatic Foods
SAPLING	Sustainable Animal Productivity for Livelihoods, Nutrition, and Gender Inclusion
SEA	Southeast Asia
SME	Small and Medium Enterprise
ToC	Theories of Change
WA	West Asia
WCA	West and Central Africa

Executive summary

By uniting livestock and aquatic food systems, Sustainable Animal and Aquatic Foods (SAAF) addresses the shared challenges of rising demand, climate change, ecosystem degradation, and food safety through integrated, context-specific innovations. Unlike siloed approaches, SAAF links productivity-enhancing, One Health, climate-smart, and digital solutions with inclusive market systems and gender-responsive strategies. The program supports smallholder producers, value chain actors, and consumers—especially women and youth, while providing policy-relevant evidence for building low-emission, nutrient-dense, and equitable food systems. Six interlinked Areas of Work drive this transformation, focusing on productivity, climate resilience, health, inclusive markets, social equity, and digital solutions.

SAAF operates in 17 priority countries, each with one or more thematic pieces of work, such as value chain strengthening or addressing cross-cutting issues like climate or health. Country coordinators play a central role in leading co-design processes and ensuring alignment with national needs. SAAF engages stakeholders across five tiers: national research systems, government agencies, research institutes, private sector, and civil society through structured co-design workshops that generate contextualized Theories of Change (ToCs). These site-level ToCs are aligned with Area of Work (AoW) pathways and serve as tools for planning, adaptation, and integration.

The SAAF ToC comprises nested AoW pathways that map from outputs to high-level outcomes, with a shared focus on scaling, policy influence, and investment. Innovation packages are tailored to specific contexts and co-developed with stakeholders to ensure ownership and uptake. During inception, several intermediate outcomes were merged for monitoring efficiency, and the 2030 outcome on policy and investment was broadened to include scaling. MELIA activities will synthesize findings from past evaluations and learning studies to guide implementation.

The inception phase sharpened SAAF's comparative advantage analysis, detailing contributions to each High-Level Outcome and priority geography. CGIAR's strengths lie in its interdisciplinary human capital, in-country infrastructure, and long-standing partnerships, enabling delivery of research with real-world impact. Twenty-eight bilateral projects from 20 donors, with a 2025 budget of USD 59 million, are aligned with SAAF's ToCs and reinforce national and stakeholder priorities.

SAAF's overall budget includes Area of Work 0 for management, MELIA, and communications. In response to a funding cut in May 2025, the program prioritized promising ongoing work and streamlined selected activities, maintaining cross-AoW integration and country responsiveness. The program is designed for flexibility, with internal reallocation mechanisms and targeted W3/bilateral funds supporting emerging priorities. Integration is reinforced through thematic areas led by AoW coordinators and guided by co-designed site-level ToCs.

SAAF is mitigating risks through adaptive management and strategic planning. Key risks include reduced Window 1 and Window 2 funding, climate shocks, disease outbreaks in animals or aquatic species, adverse policy shifts, and inflationary pressures. By maintaining responsive budgeting, strong partnerships, and flexible delivery models, SAAF is positioned to stay on course even under changing conditions.

1. Program/Accelerator overview

1.1 High-level Vision

“People’s lives and well-being in low- and middle-income countries are improved by sustainably transforming animal and aquatic food systems, so they foster inclusive, healthy and nutrient-dense food supply chains that are climate and environmentally friendly.”

Bringing animal and aquatic food systems into one program creates opportunities for broader scaling and impacts. Integration offers opportunities to share solutions across systems (mixed farming, semi-intensive, pastoral, aquaculture, and marine and freshwater fisheries). These systems face similar challenges, such as i) managing disease and food safety, ii) increasing sustainable feed resources, iii) making genetic improvements, iv) addressing environmental and climate concerns, and v) managing complex market chains. The program addresses the challenge of ensuring that both aquatic and livestock systems enable women, youth and marginalized groups to take advantage of new technical and market solutions.

1.2 Achievements from 2022–2024

The SAAF science program builds on components from four previous CGIAR Research Initiatives: Sustainable Animal Productivity for Livelihoods (SAPLING), Nutrition and Gender;

¹ Aquatic Foods;² Livestock and Climate; and One Health. Rather than simply continuing past activities, SAAF is reprioritizing at the country level and synthesizing elements from previous Initiatives into a demand driven- program.

The SAPLING Initiative worked to strengthen 13 livestock value chains in seven countries across Africa and Asia. Work on many of these value chains will continue under SAAF, but with additional emphasis on addressing issues related to One Health and Climate and the Environment. The initiative demonstrated the value of an integrated approach across work-packages to help scale solutions, combining improved productivity with market solutions and women’s empowerment. SAAF will carry forward this integrated approach as a core strategy to enhance impact and scalability.

The Aquatic Foods Initiative advanced sustainable aquatic food systems across Africa, Asia, and the Pacific by integrating innovations in breeding, digital monitoring, nutrition, and inclusive governance. The initiative showed how context-specific, science-based solutions, when linked with national systems and markets, can generate lasting impact. Under SAAF, this work will continue with enhanced focus on environmental performance, food system integration, and equity.

The One Health Initiative pilot integrated cross-sectoral approaches to health, working across seven countries. It established joint disease surveillance systems across human, livestock, and wildlife health as well as influenced reforms on food safety and antimicrobial resistance. SAAF will build upon these foundations by developing business cases and implementation guidance linked to integrated value chain approach mentioned above

The Livestock and Climate (L&C) Initiative established critical research and scaling models for climate action in livestock systems. SAAF will build specifically on the work that L&C did around climate service bundling, promoting local adaptation through an ‘adaption pioneer’ farmer led approach and improving low-emission grazing systems in Latin America, and feed and forages in Africa. The initiative’s success in influencing climate financing as well as developing monitoring

tools for resilience and emission measuring research will underpin SAAF's dual ambition: to influence systems and attract capital.

1.3 *Description of intended research*

Value proposition: By uniting livestock and aquatic systems in a single program, SAAF addresses shared pressures—rising demand, climate vulnerability, degraded ecosystems, and food safety risks—through coordinated, context-specific innovations. Unlike siloed approaches, SAAF links cutting-edge productivity enhancing, One Health, climate-smart, and digital innovations with inclusive market systems and gender-responsive strategies. SAAF science directly serves smallholder producers, other value chain actors and consumers, especially women and youth, and informs policymakers and investors seeking to scale equitable, nutrient-dense, low-emission food systems adapted to local realities.

Productivity+ focuses on increasing animal and aquatic species productivity while ensuring lowered emissions and increasing profitability for producers. Innovative research on breeding and conservation include climate-smart animal and aquatic breeding programs^{3,4} based on genetic and genomic selections;⁵ advanced phenomics and gene introgression;⁶ exploring relationships between microbiomes, feeds and emissions to select low-emission livestock; testing SLICK⁷ a heat-tolerant phenotype, in East African cattle; and proving in-vitro conservation approaches for at-risk indigenous livestock. On nutrition, examples include ingredient scoping and upgrading crop residues through biofermentation and thermochemical treatment;⁸ upskilling youth and women for market-driven- entrepreneurship in feed production and livestock fattening;⁹ screening and deployment of low-methane forages¹⁰ and *Urochloa* hybrids with greater stress tolerance.¹¹ Emerging work includes the use of animal and aquatic feeding approaches to produce fortified foods. Work on herd health includes production and deployment of new or improved vaccines, such as the heat-stable vaccine for Peste des Petits Ruminants;¹² disease risk mapping linked to climate change;¹³ and evaluating contextualized herd health packages.¹⁴

As integrating innovations across genetics, nutrition and herd and fish health is central to sustainably improving productivity, this Area of Work also considers how to incentive this, for example behavior change campaigns¹⁵ and digital tools to support decision-making. This Area of Work (AoW) also collaborates with all other AoW s to ensure that productivity enhancing innovations are integrated with social, institutional, and market solutions and are climate-smart and environmentally friendly. Research will also focus on partnerships and financial mechanisms to scale bundled Productivity+ innovations across diverse contexts.

Climate and the Environment focuses on climate-adapted inclusive solutions¹⁶ that enhance the environmental sustainability and resilience of animal and aquatic food systems. Research integrates biophysical, economic, and social science approaches to quantify the two-way impacts between AAF systems and climate change and design socio-technical solutions that will integrate AAF systems into low-emission, climate-resilient development pathways. Interventions include improved silvopastoral systems,¹⁷ integrated aquaculture,¹⁸ circular bioeconomy models, and climate-smart breeds and feeds.¹⁹ Work will advance cost-effective, context-specific tracking and monitoring tools for emissions and adaptation outcomes. This will enable producer adaptations. Scaling climate and environment innovations is enabled by mobilizing climate finance through robust evidence and developing inclusive life-cycle business models.^{20, 21}

One Health develops and supports innovations in animal and aquatic food systems to improve community health and well-being while supporting animal and environmental health.^{22, 23} Research focus includes zoonoses risks from wildlife burden of foodborne disease on health,²⁴ and risks from AMR.²⁵ In terms of innovations, we co-develop, pilot, and scale interventions to reduce and prevent zoonotic diseases and foodborne diseases^{26, 27} in traditional markers and to

reduce antimicrobial use (AMU) and mitigate AMR.^{28, 29} A growing focus is One Health operationalization, which includes exploring the costs and benefits of joint service delivery models with those of traditional single-sector service delivery methods, developing guidance for One Health implementation, supporting One Health platforms, and policy influence.

Market Systems, Policy Solutions and Scaling will provide scientific evidence, develop tools and approaches that shape market actor behaviors, influence institutional change and policy design, and catalyze increased investment. The pioneering nature of this research lies in its multi-disciplinary approach, which includes diagnosing systemic challenges, co-designing and testing innovations that catalyze beneficial participation of excluded market actors,^{30, 31, 32} and evaluating emerging changes to inform policy design and scaling.³³ Research methods include data optimization systems,³⁴ advanced analytics, participatory co-design of digital decision-support tools, institutional and behavioral studies, and analyses to inform design of business models for scaling innovations. The domain expertise required includes market systems analysis, policy analysis, data science, and institutional and behavioral analysis. We will develop and launch user-friendly platforms offering real-time data, predictive analytics, actionable recommendations, and inclusive business models leading to inclusive, efficient, and resilient AAF systems. Work on scaling is performed in collaboration with the Scaling for Impact program (see section 8.1).

Gender, Youth, and Social Inclusion (GYSI) leads strategic work and coordinates integrated work across AoW s and partners. For integrated work GYSI applies and develops frameworks on key gender considerations in technical interventions, including climate adaptation technologies³⁵ and ensures that crops and breeds respond to the needs of women and youth.³⁶ Through community-based breeding programs and inclusive service delivery, the Program tests approaches to increasing women and youth's access to services and to benefit. Strategic work focuses on empowerment and norms. We improve our tools and approaches to measure and study empowerment and create datasets, country-level status analyses and three country meta-analyses of women's empowerment. The research examines local meanings of empowerment and youth aspirations in diverse systems change.

Digital and Data Solutions will guide transitions toward more sustainable, inclusive and resilient animal and aquatic food systems. Our research will generate context-specific evidence by integrating AI-driven solutions for early warning systems³⁷ and risk management.³⁸ We will produce environmental, production, and socio-economic data to support real-time decision-making,^{39, 40} risk anticipation, and long-term planning.⁴¹ Novel approaches such as digital twins and the application of multi-scale models will combine geospatial and farm-level data to assess AAF system performance, resource use, and trade-offs.⁴² AoW 6 will inform Productivity+ (e.g., optimal feeding regimes or disease thresholds), Climate and the Environment (e.g. reducing emission or climate risk via data-driven management), One Health (e.g., forecasting zoonotic risks) and Market Systems (e.g., detecting value chain disruptions). Co-developed with national and regional partners, these tools will improve decision-making from farm to institutional levels. AoW 6 helps position CGIAR as a global leader in cutting-edge, AI-enabled research for food systems transformation, ensuring that digital innovations are scalable, gender sensitive, and tailored to support stakeholders in navigating system-level transitions.

Implementation. SAAF operates in 17 priority countries, addressing one or more thematic areas per country through integrated efforts across multiple AoW s. Examples of thematic areas are a livestock or aquaculture value chain, a pastoral or fisheries production system, or a cross-cutting issue related to animal and aquatic food production, such as One Health or climate and the environment. SAAF supports a range of animal and aquatic systems, including mixed farming, semi-intensive, pastoral, aquaculture, and marine and freshwater fisheries. For a detailed list of countries and thematic areas see [here](#).

2. Co-design and partnerships

Recognizing that transformative impact is only possible when stakeholders and partners are active, equal participants throughout the entire program cycle, we prioritize co-design, co-implementation, co-monitoring, and continuous reflection and adaptation. This ensures that priorities are set jointly, activities are collaboratively developed and carried out, and ownership of impact pathways remains with those most directly involved.

SAAF has identified priority countries along with specific thematic focus areas for each. A dedicated in-country coordinator has been appointed for each thematic area to lead and coordinate implementation. Coordinators are central to ensuring co-design by leading stakeholder and partner engagement. These coordinators will also collaborate with and link to CGIAR country coordination structures in these countries. Coordinators' local knowledge means they are well-placed to capitalize on established networks and identify new stakeholders and partners as needed.

Partners are involved in defining research agendas and priorities as well as iteratively seeking feedback through formal and informal feedback and engagement. SAAF's work with partners aims to:

- Co-design and co-implement innovations with stakeholders to ensure contextual relevance and ownership.
- Leverage complementary capacities in research, policy, and implementation.
- Facilitate scaling pathways through public-private collaboration and policy engagement.
- Promote inclusive participation of women, youth, and marginalized groups in all stages of the program.

SAAF will engage partners across five tiers

1. National Agricultural Research and Extension Systems (NARES): Core partners in co-designing and adapting innovations to local contexts. Their role includes field testing, capacity building, and policy engagement.
2. Government Ministries and Agencies: Ministries of agriculture, livestock, fisheries, health, and environment will be engaged to align SAAF activities with national priorities, facilitate policy uptake, and support regulatory frameworks.
3. Advanced Research Institutes (ARIs) and Universities: These partners will contribute cutting-edge science, training, and joint research, particularly in genomics, climate modeling, and One Health.
4. Private Sector and Market Actors: Including SMEs, tech companies, and agribusinesses, these partners will support innovation delivery, investment mobilization, and market system transformation.
5. Civil Society Organizations (CSOs), non-government organizations (NGOs), farmer facing and Community-Based Organizations (CBOs): These actors will ensure grassroots engagement, social inclusion, and culturally appropriate implementation.

For each priority country and thematic piece of work, SAAF will convene a stakeholder co-design workshop using a common framework that allows for contextualization. The output of these workshops will be co-created site-level Theories of Change (ToC) that nest logically into the SAAF AoW s and program level ToCs. Site-level ToCs will: (i) ensure a common vision and alignment of outcomes; (ii) facilitate cohesion across the AoW s; (iii) facilitate identification of context-specific

assumptions; and (iv) identify additional actors to engage. Site-level ToCs will be reviewed with stakeholders and partners on an annual basis during reflection workshops and adjustments made as needed. Further, during the annual SAAF “pause and reflect” meeting, the country coordinators will share learning lessons around co-design and partnerships, so SAAF can continually strengthen on co-design and partnerships. This approach proved highly effective in the SAPLING Initiative.^{43, 44, 45, 46}

To-date stakeholder co-design workshops to better understand demand and co-develop ToCs have been organized in Nepal, Ethiopia, and Uganda. Nepal brought together key actors from the government, private sector, farmer organizations, research and other development organizations. Priority issues to address were identified, and scaling mechanisms discussed, amongst other. In Ethiopia, multiple consultations were conducted with partners from research, extension services, and the private sector to understand their priorities and collaboratively design SAAF interventions. Efforts were also made to align with major government initiatives, leveraging these programs and projects as potential pathways for scaling. Similarly, in Uganda stakeholders were brought and discuss key constraints and opportunities, as well as learning lessons from previous initiative work. These insights will feed into participatory market studies and other interventions. The stakeholders and partners also identified new sites for scaling of proven innovation packages. In all cases ToCs were co-developed.

The remaining co-design workshops will be organized between June and August 2025. These workshops were originally planned for the inception period but were delayed pending the final budgets.

Beyond the iterative country-level process, co-design is at the heart of many research processes.

Co-design is used in participatory market systems development involving a diverse group of actors to develop a common understanding of how the animal and aquatic food market systems work. Market actors (including potential scaling partners) work to identify system blockages and collectively undertake analysis of root causes of the blockages. The shared vision informs the establishment of market interest groups around each system’s blockages; each group working with researchers to co-design and test interventions for market systems change. Actors commit themselves to taking specific concrete actions that are part of the intervention. This participatory process builds trust. Subsequent engagements are organized for actors to reflect on progress and adapt strategies.

Another example of a co-design process within SAAF research is work with adaptation pioneers. The process explicitly builds upon the aspirations, innovations and know-how of communities rather than trying to push technologies to them. It starts by meeting stakeholders individually and in small groups to understand their priorities and identify pioneer households or groups already applying effective, user-developed solutions. The aim is to define the outcomes communities want to achieve in areas like animal feed, One Health, gender equality, and climate change, and to explore how those outcomes can be reached by using their own resources and knowledge supported by research and other actors such as extension and delivery agents. Households and groups already working towards these goals will be invited to take a leading role in the follow-up work. This will lead toward an iterative learning and design process where the pioneer households take lead in facilitating change in their communities.

3. Theory of change and MELIA

3.1 *Theory of Change*

SAAF contributes to all CGIAR Impact Areas, with an emphasis on i) nutrition, health and food security; ii) poverty reduction, livelihoods and jobs; iii) climate adaptation and mitigation and iv) gender equality, Youth and social inclusion. Impact is achieved by delivering eight Program Outcomes (POs). Four are centered on innovation adoption by producers, market actors and communities, facilitated by co-design and capacity sharing. These are:

1. Animal and aquatic food producers, including women, youth and marginalized groups adopt combinations of innovations for improved productivity and profitability, emissions reduction, nutrient-dense food, healthier animals and aquatic species and climate and environmental sustainability (PO1).
2. Market actors, including women, youth and marginalized groups adopt market system innovations for equitable, efficient, low-emission and resilient animal and aquatic foods systems (PO5).
3. Communities within animal and aquatic food systems adopt innovations for increased social, economic and environmental resilience (PO3).
4. Communities within animal and aquatic food systems adopt innovations that improve community health and well-being (PO4).

The remaining outcomes center around behavior change for gender-equitable norms, increased investment, more supportive policies and integration of animal and aquatic sourced foods into the diets of producing households. Specifically:

1. Animal and aquatic food producers, market actors and communities within animal and aquatic food systems exhibit behavior supporting gender-equitable norms (PO6).
2. Public and private sector actors invest in equitable, efficient, low-emission and resilient animal and aquatic food systems (PO7).
3. Policy and decision-makers implement new or improved policies, regulations or strategies for equitable, efficient, low-emission and resilient animal and aquatic food systems (PO8).
4. Animal and aquatic food producers and their households integrate adequate safe, nutrient-dense animal and aquatic foods in diverse diets (PO2).

To achieve SAAF's vision, the program has prioritized countries and thematic areas within the countries (see [here](#)). For each of these 'site-level' ToCS will be co-developed with partners and stakeholders, including specific visions, agreed outcomes and activities. This builds on the process used in SAPLING that was both well-appreciated by stakeholders and partners, and useful in facilitating implementation (including AoW integration) and monitoring, evaluation and learning. The process also encourages the use of evidence and innovations by partners and stakeholders, leading to wider uptake and scaling.

The SAAF ToC represents a nesting of the AoW ToCs. Multiple AoW s contribute to each Program Outcome. For example, PO1 is achieved through the joint efforts of all AoW s, while PO2 is achieved through the efforts of four as seen on the ToC figures. Within each AoW ToC, impact pathways are formulated as i) outputs, which include the process of identifying and engaging with stakeholders and partners and co-design of innovations and innovation packages and capacity-sharing approaches; ii) intermediate outcomes as changes in capacities and attitudes within stakeholders and partners; and iii) outcomes as changes in the behavior of stakeholders and partners resulting from enhanced capacity (knowledge, skills, attitudes, aspirations), improved access to resources, goods and services, a more supportive and enabling environment, and shifts

in social norms among others. To enhance synergies, all AoW s share an impact pathway influencing scaling, policy and investment which, once implemented, will positively influence the enabling environment for SAAF work. Key assumptions have been attached to impact pathways. Examples include: i) capacity-sharing activities result in increased levels of knowledge and skills within groups; ii) political, economic, and governance structures and systems support investment in AAF systems innovations; and iii) decision-makers are willing to use SAAF evidence, data and tools to inform policy and investment decisions.

All innovation packages are contextualized for a specific animal and aquatic food production system (e.g. mixed farming, semi-intensive, pastoral, aquaculture, and marine and freshwater fisheries).

SAAF will work closely with partners to co-design and implement activities. The goal is to ensure partners have ownership and agency. Co-design is an iterative process that enables innovation users to have direct input into selecting, designing, re-designing, prioritizing and scaling modalities. At a programmatic level, co-design means governmental and institutional partners are central in establishing priorities, designing programs and establishing development pathways.

During the inception phase, some AoW intermediate outcomes were merged, to reduce monitoring cost and without losing the impact pathway logic. The joint 2030 AoW outcome on policies and investments was broadened to include scaling.

The SAAF ToC is connected to other Programs and Accelerators, in particular Scaling for Impact and Climate Action as detailed in section 8.1. There are also links with the three Accelerators and Policy Innovations.

We engaged with the Evaluability Assessment (EA) team by providing revised ToC diagrams. These revisions highlighted changes to the High-Level Outcomes (HLOs), Intermediate Outcomes, and 2030 Outcomes. Since the overall logic of the ToC remained consistent, we did not share the narrative from the original proposal.

The EA informal report gave the ToC an overall rating of “evaluable, pending review of supporting documentation” and affirmed the evaluability of the Area of Work ToCs, “pending review of a ToC narrative.” The SAAF SP ToC is “unevaluable” as currently formulated. We anticipate that the ToC descriptions in this report, including site-level ToCs and the MELIA studies, will be used by the EA team during their re-evaluation of SAAF.

The EA team issued six recommendations. The first two focus on revising the program-level ToC and developing a narrative that aligns with both program and AoW levels, which is now done. The third recommendation, concerning the development of country-specific ToCs, is already being implemented as detailed in both the proposal and inception report. The fourth recommendation, a follow-up to the first three and incorporating budget considerations, is also being addressed. The fifth recommendation refers to links to other Programs and Accelerators; these have been identified, primarily at the HLO level, but are not yet explicitly shown in the ToCs. Finally, the last recommendation regarding the development of evaluation questions are being addressed by reviewing the ToC assumptions in addition to research questions.

[Sustainable Animal & Aquatic Food Science Program ToC](#)

3.2 Monitoring, Evaluation, Learning, and Impact Assessment (MELIA)

For MEL, high-level outputs are tracked with PRMS, using a quality assurance system prior to entry to ensure that deliverables are mapped to the correct High-Level Output (HLO) and information is accurate. For outcomes, country and commodity level ToCs will be developed or revised, allowing the team to track outcomes and collate the evidence with partners. A standardized outcome report will be developed. Higher outcomes like policy change at regional level will be monitored by AoW teams and reported using the standard report format. Funds were allocated for monitoring outcomes in the AoW budgets.

In 2025, evaluation and learning studies initiated under the previous Initiatives will be completed through the AoW. Three adoption studies are planned, focusing on pig extension services and various forage innovations. Program and project evaluations will be finalized including women's empowerment, food safety, and One Health. Additionally, an ex-ante analysis of feed supplementation impacts on livestock productivity and a baseline study on women's empowerment in Kenya will be undertaken.

As this is the start of the Program, no new evaluation study will be done. The MELIA team will coordinate sharing the various evaluation and learning studies, to share findings to guide next steps of the research, and on methodology and implementation processes.

Up to eight causal impact assessment studies will be finalized, examining topics such as the effects of extension service delivery on animal productivity, the environmental impacts of livestock farming, and the role of aquaculture in reservoir systems. In parallel, up to three qualitative impact assessments will explore how social and gender norms within livestock value chains influence women's empowerment. These studies are led by the respective Areas of Work (AoWs).

Two new impact assessment studies will also be initiated in 2025. One will focus on designing an assessment to address SAAF's first research question, aligned with 2030 Program Outcome 1, evaluating the impact of selected innovations on the productivity or incomes of animal and aquatic food producers. The second study to be initiated will explore scaling pathways, either comparing different pathways for the same innovation or assessing the same pathway across different innovations.

[Sustainable Animal & Aquatic Foods Science Program Results Framework](#)

4. Comparative advantage analysis

The inception phase added granularity to the comparative advantage analysis with details for each HLO, reviews of specific parties, and a finer geographical scope consistent with the PORB and prioritization exercise.

CGIAR's comparative advantage lies in its diverse and interdisciplinary human capital, in-country presence and facilities, extensive social capital, and strong incentives that drive impactful research and innovation across Areas of Work.

Human Capital. Across all AoW s, CGIAR draws on a specialized pool of scientists and analysts. Productivity+, Climate & Environment, and GYSI benefit from strong teams of biophysical, social, and data scientists. One Health brings interdisciplinary expertise in public health, water quality, and policy. Market Systems and Digital & Data emphasize economists, IT solution architects, policy analysts, and gender researchers.

Biophysical Capital. All AoW s leverage CGIAR's robust biophysical infrastructure. This includes laboratories equipped for animal health, GHG analysis, and aquaculture research; animal handling facilities; secure data storage and computing infrastructure; and in-country research hubs that support local implementation.

Social Capital. CGIAR's long-standing partnerships underpin its success across all AoW s. These include relationships with national R4D organizations, national agricultural research and extension systems (NARES), universities, NGOs, and private sector actors. One Health and Market Systems benefit from particularly broad engagement with government and policy actors, while Digital & Data collaborates with global and local data networks.

Across six AoW s and 22 HLOs, we reviewed the comparative advantage of 165 organizations.

11% have been rated as **highly efficient and impactful** in delivering the specific HLO. These include NARES such as the National Institute of Veterinary Research in Vietnam or the Tanzania Livestock Research Institute; the private sector, including AbacusBio or Kuza Biashara; universities, such as Wageningen. Though highly efficient, these organizations have a relatively narrow scope or geographical outreach, and hence there might be mutual interest for collaboration and wider impact to be considered case by case.

9% have been rated as **minimally efficient and impactful** for delivering SAAF outputs and outcomes. These organizations are either think tanks and consultancy firms (e.g. MacKenzie), governmental extension services (Fisheries Commission Ghana), or NGOs (Heifer International, CARE). In those instances, CGIAR is well-positioned to undertake the work. Still, stakeholder engagement can be pursued with stakeholders to transition toward sustainability and scaling.

The remaining 80% have been rated as **moderately efficient and impactful** in undertaking HLO work. This group encompasses NARES (KALRO, IFRDI), NGOs (Vétérinaires Sans Frontières); private sector organizations (EON Aquaculture Limited, Githunguri Dairy Cooperative); international organizations (FAO, World Bank, AfDB); universities from the Global North (Corwell, Kiel) or targeted countries (Universidad de los Andes, University of Nairobi, Hanoi University of Public Health). While CGIAR is well or moderately positioned in most instances, there are significant opportunities for collaboration, stakeholder engagement, and in some instances, formal partnerships.

[Sustainable Animal & Aquatic Food Science Program Comparative Advantage analysis](#)

5. Prioritization

5.1 *Knowledge-driven priorities*

Prioritization results (here) suggest that SAAF's highest impact potential across all regions and AoW s is for food security and poverty reduction, followed by climate adaptation and mitigation and social inclusion, with the least impact on environmental health.

South Asia (SA) and East and Southern Africa (ESA) emerged as regions with the greatest potential for impact across multiple CGIAR Impact Areas, followed by West and Central Africa (WCA), Southeast Asia and the Pacific (SEA), and Latin America and the Caribbean (LAC). Central and West Asia and North Africa (CANA) showed the lowest overall potential impact.

There is strong potential for food security impact in SA and ESA and to a lesser extent in WA; for poverty reduction in ESA and WA, with more moderate potential in SA; and for climate adaptation and mitigation in LAC, SA, and SEA. Moderate impact potential was also indicated for social inclusion in WA, ESA, and SA and for environmental health in LAC, SA, and SEA (Table 5.1).

From an Area of Work (AoW) perspective, Productivity+ primarily contributes to food security and poverty reduction, with secondary contributions to social inclusion and climate outcomes. Climate and the Environment, along with Digital and Data Solutions, are key drivers of climate adaptation and mitigation and improved environmental health. One Health contributes significantly to both food security and environmental health, while Market Systems, Policy Solutions, and Scaling play a major role in advancing food security and reducing poverty. Finally, Gender, Youth, and Social Inclusion contributes most strongly to social inclusion, followed by poverty reduction. Though expected, this result from the prioritization confirms the complementarity of the AoW to deliver across the different impact areas.

Based on prioritization scores alone, the largest budget shares would be expected in ESA and SA, moderate allocations in WA, SEA, and LAC, and minimal allocations in CANA. The overall distribution of 2025 budgets shows both alignment and divergence from these expectations. Across all AoW s, ESA and SA received 45% and 11% of the total regional budget respectively. SA being notably underfunded relative to its high prioritization score. WA, SEA, and LAC received 9%, 14%, and 2%, respectively, with LAC receiving less than expected for a moderately prioritized region. CANA received 1%, consistent with its lower impact potential. Additional factors underpinning SAAF's final decision-making on prioritization are given below.

Table 5.1. Prioritization results by impact area and region across all SAAF high-level outputs.

Impact area	Region					
	Latin America and the Caribbean	Central and West Asia and North Africa	West and Central Africa	East and Southern Africa	South Asia	Southeast Asia and the Pacific
Food security						
Poverty reduction						
Social inclusion						

Climate adaptation & mitigation						
Environmental health						

Note: Prioritization scores averaged across all SAAF high-level outputs. Color code: average scores 0.54 to 0.40 are dark green; 0.39 to 0.30 light green; 0.29 to 0.20 dark orange; 0.19 to 0.10 light orange; 0.09 to 0.04 no color

5.2 *Final decisions on the Program/Accelerator's direction and design*

Across all AoW and HLOs, funding is shaped not only by prioritization scores but also by additional factors, including local demand, enabling environment, and readiness to scale, as well as SAAF's overall approach of interlinkages between the AoW to produce outcomes (and thus impact).

Productivity+ ESA consistently receives the largest share of funding across HLOs (46–57%), reflecting its high prioritization scores in poverty reduction, food security, and social inclusion, strong ongoing programs (particularly in genetics and feeds/forages), and the presence of proven innovations ready for scaling. SA, SEA, and WA receive more moderate allocations. SA scored highly on prioritization but currently has limited active programming. SEA demonstrates moderate impact potential and supports ongoing Initiatives such as GIFT tilapia and forage innovation. WA's allocation is driven by promising new innovations (e.g., the thermotolerant Peste des Petits Ruminants vaccine) and efforts to adapt and scale successful models from ESA. LAC and CANA receive little to no budget, due to low prioritization scores (CANA) and only moderate comparative advantage (LAC). Opportunities to expand Productivity+ engagement in SA and LAC, where strategic potential exists, will be explored in future phases.

Climate and the Environment Climate and the Environment funding reflects regional impact potential, synergies, and scalability. ESA receives 12–43% across HLOs, driven by strong alignment with Productivity+ and innovation pipelines. SA holds 48% of the climate finance HLO, linked to dense systems and de-risking tools. LAC receives 9% and 35% in two HLOs, focused on biodiversity and low-carbon solutions. This AoW balances regional opportunities, partner demand, and funder priorities: ESA and WA emphasize adaptation; LAC, mitigation; SA and SEA, innovation bundling; and CANA, targeted aquaculture solutions. WA receives less than its potential due to feasibility limits, while CANA is selectively included. A strong global focus (45%) supports tool development and cross-regional evidence in response to growing climate threats.

One Health ESA receives the largest share across most HLOs (upwards from 44%) consistent with its high impact potential, strong existing programs, and readiness to operationalize One Health approaches. SEA receives moderate allocations (22–26%, barring one HLO), building on long-standing institutional partnerships, and alignment with national and regional priorities. SA, with high impact potential, also receives moderate allocations (25% into two HLO), largely due to fewer ongoing programs and limited scaling opportunities. WA and LAC are deprioritized in most HLOs due to limited existing activities to build upon.

Market Systems, Policy Solutions and Scaling ESA receives the largest share of regional funding across most HLOs (36–73%), reflecting strong impact potential, active partnerships, and a pipeline of scalable innovations. SA receives moderate funding despite high prioritization scores, due to limited ongoing programs and delivery infrastructure, though investments and policy target enabling tools and policy engagement. SEA also receives moderate allocations, based on lower prior engagement. WA receives modest funding under one HLO aligned with regional demand but is deprioritized elsewhere due to institutional constraints. LAC and CANA receive minimal or no funding, though LAC shows some potential. Up to 85% of funding in some HLOs is directed to

global activities, emphasizing the role of upstream research, digital tools, and coordination in driving impact.

Gender, Youth and Social Inclusion ESA consistently receives the largest share of funding (23% to 83%), reflecting strong impact potential, established programming, and alignment with other SAAF Areas of Work. SEA and LAC are engaged through targeted One Health and forages, aligned with impact potential on climate and the environment. SA and WCA receive focused investments in response to country-level demand. Global allocations (up to 11%) support cross-regional tool development and evidence generation.

Digital and Data Solutions ESA receives the highest investment (18%, 24% for two HLOs), justified by strong prioritization, mature ecosystems, and readiness in countries like Kenya, Zambia, and Ethiopia. SA receives modest but targeted funding (4%), leveraging its role as a testbed for FAIR data systems in aquaculture. SEA, WCA and CWANA received no funding, largely due to limited digital infrastructure and institutional readiness, a lack of comparative advantage (SEA), and gaps in enabling environment and institutional capacity (WCA and CWANA). A significant share of funding (76–100%) goes to global efforts, building core AI tools, data standards, and digital platforms for future regional uptake and system-wide interoperability.

Moving forward SAAF will explore further engagement in SA and LAC considering local impact potential, partner and stakeholder demand, and comparative advantage.

6. Alignment of W3 and bilaterally funded work

Twenty-eight mapped bilateral projects from 20 donors are supporting SAAF in advancing its vision, with a total 2025 envelope of USD 59 million. Most bilaterals were assessed as highly complementary and all align well with SAAF's Theories of Change (ToCs) and support partner and stakeholder priorities. Examples include anti-methanogenic feed research for low-emission livestock systems; integrated multi-trophic aquaculture in Asia and Africa addressing sustainability, health, and nutrition; and One Health capacity-building in Eastern and Southern Africa focused on surveillance, biosecurity, and system integration.

Productivity+ receives the largest share of bilateral funding (41%), supporting both animal and aquatic food systems across its three core pillars: breeding, health, and nutrition, at both upstream and applied levels. However, no bilateral currently supports its emerging work on food fortification. One Health also receives a significant share (33%), with strong investment in operationalizing One Health approaches. Climate and the Environment receives moderate funding (15%), primarily focused on climate-adapted innovations, resilience-building, and evidence generation. The Market Systems, Policy Solutions, and Scaling Area of Work receives 8% of bilateral funding, supporting activities across aquatic and livestock markets and spanning all three thematic components. Gender, Youth and Social Inclusion (1%) and Digital and Data Solutions (2%) have limited bilateral investment. GYSI-focused bilaterals engage with both livestock and aquatic systems, emphasizing women's and youth empowerment, while the digital bilateral supports tools for climate-smart livestock systems. Overall, 17 of SAAF's 23 HLOs are supported through bilaterals. Geographically, the bilaterals are in eight ESA countries. There are one or two countries each in of WCA, SEA, SA, and LAC.

While all bilaterals contribute to SAAF's innovation portfolio and learning, notable gaps remain in scaling, Gender, Youth and Social Inclusion, Digital and Data Solutions, and in regional coverage beyond ESA, highlighting priorities for future resource mobilization.

7. Plan of Results and Budget

SAAF's total budget for 2025 is USD 20 million, with 17% allocated to management, MELIA, and communication and knowledge management. Among the Areas of Work (AoWs), Productivity+ receives the largest share (40%), reflecting its broad scope across genetics and breeds, nutrition, and animal health. This is followed by One Health (15%), Market Systems, Policy Solutions and Scaling (10%), Climate and the Environment (9%), Gender, Youth and Social Inclusion (6%), and Digital and Data Solutions (3%). As a newly established AoW, Digital and Data Solutions will be a priority for future resource mobilization.

In response to the May 2025 budget reduction, the program retained the highest-priority activities as identified through the knowledge-based prioritization process and other considerations (see Section 5). SAAF remains flexible to reallocate funding based on emerging needs and opportunities. Program integration is actively supported through multiple mechanisms, including cross-AoW contributions to country level thematic work, in-country coordinators facilitating integration, and dedicated budgets for stakeholder co-design workshops where site-level Theories of Change (ToCs) will be developed to guide and monitor integrated delivery.

Productivity+ allocates 83% and 17% to outputs and outcome evidence generation, respectively. The largest share of the budget (38%) is allocated to OP1.1 on breeding and conservation, with a strong emphasis on scaling proven breeding programs and building partner capacity to sustain them beyond SAAF's direct involvement. OP1.2 (feeds and forages) receives 22%, reflecting the critical role of nutrition, often the highest production cost, in both animal and aquatic food systems. OP1.3 (health and welfare) receive 11%, complementing investments under the One Health AoW. OP1.5 (innovation integration) also receives 11%; this newly introduced budget line responds to partner demand for bundled solutions that address multiple productivity constraints. OP1.4 (food fortification) is an emerging area with a small budget for scoping work: budget to this output may expand in later years depending on impact potential. The outcome budget supports evidence collection against all Productivity+ outcomes and two program-level outcomes. Most of the Productivity+ budget (86%) is allocated to in-country activities across all 17 SAAF countries, while the remaining 14% supports global work, including upstream technology development. ILRI provides 64% of the Productivity+ budget, followed by WorldFish (19%), ABC (9%), and ICARDA (8%). ILRI and ICARDA support work on livestock species, WorldFish on aquatic species, and CIAT contributes through forage development.

Climate and Environment allocates 99.3% of its budget to outputs and 0.7% to outcome-level evidence generation, supporting both near-term innovation delivery and long-term environmental impact. The largest share (58%) funds OP2.1 on emissions mitigation, focused on piloting and scaling low-emission practices such as improved feeds, forages, and genetics. OP2.2 (8%) targets resilience and resource-use efficiency, while OP2.3 (26%) supports biodiversity-positive, ecosystem-based approaches. OP2.4 (7%) is dedicated to measuring, reporting and verification (MRV) systems aligned with climate finance and national reporting needs. Approximately 82% of the AoW 2 budget is allocated to in-country work across all SAAF countries; the remaining 18% supports global analytics, upstream innovation, and policy engagement. ILRI manages 28% of the budget, drawing on expertise in livestock emissions, MRV, and integrated resilience. WorldFish contributes 5% with a focus on climate-smart aquaculture systems; IWMI manages 12%. CIAT brings 39% through its strengths in climate-resilient forages and landscape restoration. Bioversity contributes 9%, while ICARDA manages 7% supporting integrated dryland strategies and rangeland management.

The **One Health** AoW invests in reducing health risks at the interface of animals, humans, and the environment. Of the total budget, 81% is directed to outputs and 19% to outcome evidence generation. OP3.1 -zoonotic disease surveillance and control - receives 28% of the AoW budget, reflecting the importance of early detection systems and national policy support to manage zoonoses in livestock and aquatic food systems. OP3.2, food safety in informal markets, is allocated 31% recognizing the high disease burden associated with unsafe food in traditional markets and the opportunity for low-cost, high-impact interventions. OP3.3 antimicrobial resistance (AMR) mitigation and responsible antimicrobial use receives 33%, given the rising global concern around AMR and the need for evidence-based strategies to reduce antimicrobial use in semi-intensive aquaculture and livestock systems. OP3.4 Operationalization of One Health, a newer thematic area, is allocated 7%, supporting grassroots capacity, service delivery models, and evaluations to improve implementation efficiency. The majority of budget (96%) goes to SAAF priority countries, with 4% supporting global work. The work is supported by ILRI (71%), IWMI (14%), IFPRI (9%), and WorldFish (6%).

Market Systems, Policy Solutions and Scaling allocates 68% of its budget to outputs and 32% to generating outcome evidence. The largest share (28%) is directed to OP4.3 on institutional and behavioral models for innovation uptake, focusing on business models for bundled innovation delivery and associated adoption and impact assessments. OP4.1 (information systems) and OP4.4 (policy analysis and engagement) each receive 15%, reflecting their complementary roles in enhancing animal and aquatic food systems through micro- and macro-level innovations. OP4.2 (decision support tools) is allocated 5%, and OP4.5 (scaling strategies) receives 4%, with additional contributions expected from the Scaling for Impact program as scaling pathways mature. Of the total budget, 73% supports in-country activities across 12 SAAF countries, while 27% is allocated to global work, particularly for developing policy and market analysis tools. ILRI provides the majority of funding (57%), followed by IWMI (25%), WorldFish (7%), ICARDA (7%), and ABC (4%).

Gender, Youth and Social Inclusion allocates 64% of their budget to outputs and 36% to outcome generation. OP5.1 (tools and datasets) and OP5.2 (women and youth empowerment and equitable norms and policies) receive 31% and 29% of the budget, respectively. OP5.3 (integrated gender) receives only 5% of the GYSI budget, given additional funding for this is available through the other AoW. 94% of the budget supports country work and 6% global work. ILRI Provides 70% of the GYSI budget, followed by WF (11%), ICARDA (10%) and IWMI and CIAT (5% and 4%, respectively).

Digital and Data Solutions prioritizes strategic investment in digital public goods, AI-enabled tools, and FAIR data systems to advance climate-smart, inclusive animal and aquatic food systems. The largest share (41%) supports OP6.2, reflecting strong demand for exploratory digital solutions and decision support tools. OP6.1 and OP6.2 receive 26% and 33%, respectively. Here priority has been given to joint deliverables with other Science Programs/Accelerators, such as ShambaShield (Climate Action) and the Omni Financial Advisor app (Digital Transformation Accelerator). The PORB addresses emerging priorities by expanding from aquaculture productivity to broader aquatic system governance and sustainability metrics. With 83% of the AoW 6 budget invested in global public goods, the portfolio enables integration across programs, platforms, and geographies. AoW 6 is pivotal to addressing persistent gaps in digital infrastructure, literacy, and inclusive decision-support systems. This work is supported by WF (66%), ILRI (22%), and CIAT (12%).

[Sustainable Animal and Aquatic Foods Science Program 2025 Plan of Results and Budget](#)

8. Cross-Portfolio linkages and geographic coordination

8.1 Cross-portfolio linkages

SAAF has a number of cross-portfolio linkages, of which two examples are given below.

SAAF will work with S4I program to co-develop tools for assessing scaling readiness of innovations and use these to prioritize innovations for scaling. This process is enabled by an embedded scaling expert from S4I with whom SAAF works to optimize innovations, beginning with the ones that have been in the scaling pipeline under the initiative and are matured for scaling success. Working with the embedded scaling expert and through the scaling fund, SAAF will define and refine scaling pathways for “*fish cage production in small reservoirs in Ghana*” and “*an embedded private sector extension approach – dairy farmer advisor (DFA) in Kenya*” and document lessons that can continuously improve the scaling of emerging SAAF innovations while contributing to the adaptive learning agenda for S4I. The evaluation of different scaling pathways, including through rigorous approaches, will contribute to the SAAF impact assessment agenda. For innovations under development, the embedded scaling expert will be involved at the early stages including through market systems diagnosis processes that involve potential scaling partners and which SAAF leverages to co-design innovations that match the interest of scaling partners, therefore shortening the route to scale. Moreover, the market system work under SAAF, especially on innovative financing will enhance adoption and scaling of SAAF innovations while contributing to the S4I initiative on unlocking financing for scaling.

SAAF and the Climate Action Science Program (CASP) are aligned in reducing greenhouse gas emissions and building resilient agrifood systems. CASP generates system-wide tools, evidence, innovations, and policies, while SAAF delivers contextualized solutions for animal and aquatic foods. Their collaboration enables two-way learning across scales—SAAF’s work on silvopastoral systems, integrated aquaculture, and climate-resilient breeds informs CASP’s low-emission strategies and MRV systems. Joint tools like life-cycle assessments, remote sensing, and digital MRV support smallholder adoption and carbon market access. Together, they advance climate risk management solutions, as well as climate finance and policy through shared platforms, co-develop briefs, and investment frameworks, contributing to scaled low-emission practices and the mobilization of climate finance for AAF systems. Shared personnel and data systems further reinforce this integration.

8.2 Geographic coordination

SAAF works in countries where there are multiple science programs, requiring geographic co-ordination. As an example, in Vietnam there is SAAF, Better Diets and Nutrition (BDN), Multifunctional Landscapes (MFL), Breeding for Tomorrow (B4T), and Scaling for Impact. By co-locating activities, aligning stakeholder engagement, and integrating program delivery with national food system transformation priorities. For example, SAAF plans to coordinate in Northwest Vietnam with BDN on the intersection of diets, nutrition, and food safety, and with MFL to jointly address agroecology and livestock development among ethnic minority communities. In the Mekong Delta, SAAF and BDN can collaborate on aquatic food systems, while B4T will focus on climate-adapted genetic innovations for rice. These coordinated efforts aim to enhance synergy, improve delivery efficiency, and ensure responsiveness to national priorities, including Vietnam’s vision for a regional food innovation hub in Southeast Asia.

9. Risk management

SAAF is mitigating risks through adaptive management, strategic partnerships, responsive budgeting, and targeted innovation to maintain progress under changing conditions. The most critical risks are:

- i. Reduced Window 1 and 2 funding due to shifting donor priorities and reduced ODA;
- ii. Epidemics affecting animal or aquatic species, both of which could significantly lower the number of farming and fishing households reached;
- iii. Economic downturns and inflation threaten demand for nutrient-rich animal and aquatic foods, reducing producers' incomes and ability to invest in innovations;
- iv. Adverse national policy shifts regarding inclusion and equality; and
- v. Increased frequency of extreme climate events, which could undermine social progress and resilience.

Please refer to the following [link](#) with the Program/Accelerator's full risk register. This link is open to ISDC colleagues and others will be granted access, on request.

Category	Risk	Description	Current Risk Level	Target Risk Level	Actions to Manage Risk
Funding	Reduced Window 1 & 2 Funds	Reduced Window 1 & 2 funds caused by political changes from national governments which donate the bulk of CGIAR funds or other changes in donor priorities. This could result in outcome achievements failing to reach the anticipated targets for the number of farmer and fisher households reached by SAAF.	16	9	Advocacy and engagement for funding on animal and aquatic food systems. Use of adaptive management to update outcome targets in line with budget reduction.
Business continuity	SAAF species epidemic	A significant reduction in production of one or more animal or aquatic food species caused by an epidemic. This would result in reduced production and profitability for farmers and fishers, less nutrient-dense food, sicker animals and aquatic species and damage to climate and environmental resilience, that would impact the number of farmer and fisher households reached with SAAF innovations.	15	9	Strong partnerships to incentivize early information sharing on epidemics; quick reaction to work with partners on quantifying the impact of the epidemic for future learning; revise activities and budget allocation as needed.
Business continuity	Economic downturns and Inflationary pressures	Reduced purchasing power and declining or stagnating demand for animal and aquatic sourced foods because of economic	9	4	Prioritize innovations that significantly increase affordability of animal and aquatic foods to substantially cushion

		downturns and rising inflation in lower- and middle-income countries. This may lead to reduced income for animal and aquatic food producers and hence less investment in innovations and increasing vulnerability of animal and aquatic-species dependent livelihoods.			demand from inflationary influences.
Partners and partnerships	Anti-inclusion and equality initiatives	A substantial change in national policies around inclusion and equality was brought about in a SAAF priority country. This could result in a society unwilling or unable to modify existing social norms and less progress toward achieving inclusion and equality in animal and aquatic food systems.	9	4	Strong partnerships with decision-makers on policy related to inclusion and equality, and animal and aquatic food systems, for advocacy and to be conversant of the policy changes. Quantify impacts of policy change for future learning and advocacy; revise activities and budget allocations as needed.
Business continuity	Impacts of Climate Change	Droughts, forest fires, floods and other environmental events which damage or destroy farms and fisheries occur more frequently due to climate change. One event could halt or reverse progress toward reaching intended number of farmers, fishers and institutional partners adopting SAAF innovations.	9	6	Build resilience of communities; quantify impacts of events for future learning; revise activities and budget allocations as needed.

10. Addressing feedback on select topics

Concise summary of feedback	Details on how the Program/ Accelerator has addressed or will address this feedback (in the Inception Phase or beyond)
There is a lack of information on how water systems are covered in the Portfolio.	Water systems are a strong focus of the SAAF Portfolio, addressed through climate-smart innovations and in-country case studies that promote climate risk management, efficient use and circular bioeconomy strategies. The Climate and Environment component explores interactions between water resources and animal and aquatic food systems, including the impacts of climate change on water availability and quality. Solutions include integrated aquaculture, recirculating water technologies, and silvopastoral practices that enhance efficiency and ecosystem resilience. SAAF also works with national and sub-national institutions to support inclusive water governance and community engagement in sustainable water management.
There is a lack of information on how climate mitigation is covered in the Portfolio.	Climate mitigation is a priority in the SAAF Portfolio, integrated across research, innovation, and policy components to reduce greenhouse gas emissions from animal and aquatic food systems. The program advances bundled low-emission solutions such as improved breeds and forages, sustainable feeds, and nature-based livestock and aquaculture systems that enhance productivity while lowering environmental impacts. To support uptake, SAAF promotes productivity enhancing innovation bundles that are also profitable as well as climate finance mechanisms (including credit scoring, index-based insurance, and results-based financing) that incentivize adoption of low-emission practices. It also supports community-based resource management to increase ecosystem carbon storage and works with governments to align policies with national and global mitigation targets, including nationally determined contributions and long-term climate strategies. These combined efforts aim to deliver measurable climate benefits while enhancing resilience and livelihoods.
There is a lack of information on how trade-offs and synergies in mixed systems are covered in the Portfolio.	The first of SAAF's program-level research questions focuses on the trade-offs and synergies across productivity, profitability, emissions reduction, resilience, One Health, and equity in innovation packages for producing safe, nutritious animal and aquatic foods—most of which come from mixed systems. This is addressed through several AoW -level HLOs, such as HLO6 of AoW 1 (adoption of innovation bundles for productivity, profitability, and emissions reduction) and HLO1 of AoW 2 (evidence on the impacts of AAF systems on climate and environment). Mixed systems are also central to collaborations with CGIAR Initiatives like Sustainable Farming and Multifunctional Landscapes, where we will jointly examine dietary trade-offs and how policy and market innovations shape outcomes.
There is a lack of information on how environmental health and biodiversity is covered in the Portfolio.	Environmental health is a focus of the Portfolio, addressed through both the One Health and Climate and Environment AoW s. One Health addresses the interconnections between people, animals, plants, and their shared ecosystems. One Health reducing risks from zoonotic diseases, antimicrobial resistance and foodborne illnesses, while actively safeguarding environmental health The Climate and Environment AoW contribute to environmental health by accelerating innovations—such as improved breeds, feeds, and management practices—that reduce greenhouse gas emissions, enhance resource efficiency, and minimize pollution from animal and aquatic food systems. These innovations promote more sustainable production landscapes, helping to preserve ecosystems, soil health, and water quality.

Concise summary of feedback	Details on how the Program/ Accelerator has addressed or will address this feedback (in the Inception Phase or beyond)
	Biodiversity is addressed through research in in-vitro livestock conservation approaches, forage breeding, feed systems, and sustainable aquaculture practices that contribute to ecological balance and resilience.
There is a lack of information on how youth and social inclusion (that does not relate to gender) is covered in the Portfolio.	Youth considerations are embedded in all six SAAF Areas of Work: • Productivity+: Ensures youth access to innovations in breeding, feeding, and health. • Climate and Environment: Engages youth in climate-smart practices and green jobs. • One Health: Involves youth in food safety, antimicrobial resistance awareness, and health service delivery. • Market Systems and Policy: Promotes youth participation in inclusive market models and policy dialogues. • Digital and Data Solutions: Addresses the digital divide and supports youth-led tech innovation. The GYSI AoW looks at the context-specific nature of youth and gender to understand how interventions are being used by youth from diverse backgrounds or ethnic groups. Applying a gender lens is often the first step in unpacking these intersectional dynamics.
There is a lack of information on how capacity sharing (as it relates to internal learning and decolonization of research) is covered in the Portfolio.	Capacity sharing is embedded in the program Theory of Change, and each Area of Work has different ways to carry this out. For example, a major component of Productivity+ is working with national centers of excellence in genetic breeding and livestock and aquatic production to strengthen the use of advanced genomic tools in breeding-related work. This AoW also works on disease diagnostics, biosecurity, and awareness raising on access and benefit sharing of genetic resources South–South exchange and triangular cooperation is a main component of these efforts. Additionally, One Health focuses on building institutional capacity to strengthen integrated One Health networks within countries for more coordinated responses to outbreaks and disease control. It will co-generate data and knowledge management platforms for national partners, particularly national research and meteorological institutes. SAAF takes concrete steps to decolonize research by prioritizing locally defined challenges and solutions in research design and delivery; embedding community-based approaches and indigenous knowledge in innovation development; creating locally appropriate delivery systems for innovations, rather than imposing externally developed models; and ensuring knowledge sovereignty through open genomic databases and co-developed digital platforms that respect FAIR and inclusive data principles.

Appendix 1: Addressing specific feedback

Concise summary of feedback	Details on how the Program/Accelerator has or will address this feedback (in the Inception Phase or beyond)
Define geographic focus as well as species/varieties	SAAF has 17 priority countries, spanning all CGIAR regions. The focus of work in each country is well-defined, as well as the SAAF AoW contributions to each. SAAF targets a range of livestock species, including chicken, pigs, small ruminants, cattle (beef, dairy and dual-purpose), and buffalo, as well as aquatic tilapia and carp species. The target animal and aquatic species were selected based on low- and middle-income country demand for animal and aquatic source foods.
Plan for enhanced AoW integration	SAAF has prioritized work that integrates across multiple AoW s, depending on demand. For example, work on strengthening the carp-polyculture systems in Bangladesh, and the beef cattle and pig value chains in Vietnam each engage five AoW s. To support AoW integration, site-level Theories of Change will be co-developed with stakeholders and partners for all SAAF's thematic pieces of work (which number around 30, in 17 countries). These will show the specific contributions of, and integration between, the different AoW components, and how these jointly deliver on outcomes.
Revise risk management	See Section 9
Revise comparative advantage analysis	See Section 4
Develop full MELIA plan	See Section 3
Plan for operationalization of integrated gender and social inclusion	The Gender Youth and Social Inclusion (GYSI) AoW leads and holds the budget for all strategic GYSI research, conducting and reporting on this work. Integrated GYSI research is funded by other AoW s and carried out collaboratively, with GYSI scientists working closely with thematic teams. Our approach uses gender and youth as primary entry points, applying an intersectional lens that considers how additional factors—such as ethnicity, caste, marital or social status, and education—shape marginalization. The team has strong gender expertise and growing youth capacity, and partners with specialists to address areas, such as disability, trauma, or refugee status. AoW 5 also collaborates closely with Gender Equality and Inclusion accelerator.
Modalities for context-specific and effective scaling strategies	The SAAF program will roll out a structured dissemination and scaling plan, leveraging regional partnerships, digital platforms, and collaboration with the Scaling for Impact Program. Using jointly developed criteria, SAAF has already identified the top 10 “innovations in use” to prioritize for scaling support. A market systems-based approach also underpins scaling, ensuring scaling efforts are grounded in an understanding of systemic constraints, incentives, and actor roles. SAAF actively involves market actors in participatory market system diagnostics and in co-designing and testing interventions. This engagement enables lead actors to adopt and expand emerging solutions, while the research team tracks and analyzes scaling pathways to inform replication and adaptation across regions.
Implementation of co-design processes	See Section 2

Contextualization of research and alignment with regional frameworks	SAAF research is based on regional priorities in Asia, Africa, and Latin America. In Africa, African Union Inter bureau of Animal Resources (AU-IBAR) is a close partner in setting up regional information systems and policy advocacy, and SAAF also works with the Comprehensive Africa Agriculture Development Program (CAADP). In Asia, SAAF works through Association of Southeast Asian Nations (ASEAN) and other regional bodies such as the Mekong River Commission. SAAF will also engage in regional fora to share its research such as the African Food Systems Forum.
Ensuring environmental degradation & ecological challenges are addressed	Environmental degradation and ecological challenges will be addressed through the Climate and Environment AoW. This includes tailoring research to local contexts and evaluating how different animal and aquatic food systems impact the environment, and tailoring cost-effective strategies to reduce emissions, rehabilitate resources, and mitigate risks.
Compliance with Nagoya protocol and access and benefit sharing	The SAAF program will comply with the Nagoya protocol and its principles on access and benefit sharing. Within the SAAF partner CGIAR institutes, particularly the International Livestock Research Institute and WorldFish, there significant expertise on access and benefit sharing through the legal and research compliance departments. SAAF will leverage this expertise to ensure compliance with national ABS laws, including on prior informed consent and mutually agreed terms. Where there are no national laws, SAAF will use agreements of a similar nature.
Market systems approach	Innovation will follow a market systems diagnostic phase, aligned with the research-for-development pathway to support scaling and implementation. Market-based solutions will be co-designed and tested, with scaling efforts informed by diagnostic work already initiated under earlier Initiatives and bilateral projects. This approach is tailored to specific value chains and geographies, grounded in stakeholder engagement and evidence generation. In areas where land tenure presents barriers, the program will assess the enabling environment and explore inclusive models—such as cooperative arrangements, integrated land and water use, and community-based management of pastoral lands and aquaculture ponds—to support sustainable animal and aquatic food markets.
Centralized data management framework	The Digital and Data systems AoW serves as the integrative backbone to integrate data systems across work areas for enhanced coherence. AoW 6 will establish a cross-AoW data governance framework rooted in FAIR principles, prioritizing interoperability and stakeholder needs to harmonize protocols. The aim is not to ‘reinvent the wheel’ but to standardize data collection across Productivity+ (e.g., genetic data from Ethiopia’s small ruminants), Climate and Environment (e.g., GHG emissions in Colombia’s livestock sector), and One Health (e.g., Uganda’s zoonotic surveillance) via shared taxonomies and then seek integration through context- and user-centric dashboards. AoW 6 will build shared data hubs to integrate, for example, markets data and gender and social inclusion insights to enable cross-sector analysis.
Implementation of structured dissemination plan to scale research outputs	To scale the visibility and uptake of SAAF’s research outputs, the communications team is developing a multi-pronged communication strategy designed to align with both global and national priorities, while leveraging the strengths of participating centers. This structured dissemination plan ensures that research findings are strategically positioned to drive adoption, influence policy, and engage diverse stakeholders across priority countries. At the global level, SAAF will develop high-impact messaging aligned with CGIAR’s impact areas. A dynamic dissemination system will repackage research

	results and outcomes into accessible formats including an e-letter, integrated social media campaigns, outcome stories, policy briefs, explainer videos and media outreach. Additionally, in collaboration with partners, SAAF will develop a regular webinar series to amplify innovations and emerging evidence to global audiences. At the national level, SAAF will implement a structured communication approach by drawing upon communication support from centers to co-design country-specific dissemination strategies. These strategies will respond to the unique needs of national stakeholders and audiences, ensuring that emerging research findings are translated into accessible, relevant formats. Dissemination will be carried out through local media (TV, radio, community platforms) local social media channels and materials will be produced in local languages where appropriate.
Addressing sustainability issues, especially crop-livestock-aquatic system integration and pastoral resource management	Multiple Areas of Work within SAAF will address sustainability, including Climate and the Environment—working alongside Productivity+ to ensure productivity-enhancing innovations are environmentally sustainable—and Markets, Policy Solutions, and Scaling, which focuses on engaging the private sector and fostering supportive policy environments for sustainable innovation delivery. SAAF will continue to address resource management in different systems, including mixed and pastoral systems, drawing from previous CGIAR Initiatives including on circular approaches. During the inception phase, SAAF strengthened collaboration with related programs such as Sustainable Farming, Multifunctional Landscapes, and Climate Action to ensure a comprehensive approach that considers all critical aspects of diverse farming systems.

-
- ¹ CGIAR Initiative on Sustainable Animal Productivity. 2025. CGIAR Research Initiative on Sustainable Animal Productivity: Annual Technical Report 2024. Montpellier, France: CGIAR System Organization. <https://hdl.handle.net/10568/174144>
- ² CGIAR Initiative on Aquatic Foods. 2025. CGIAR Research Initiative on Aquatic Foods: Annual Technical Report 2024. Montpellier, France: CGIAR System Organization. <https://hdl.handle.net/10568/174128>
- ³ Tolemariam, T., Abegaz, S., Getachew, T., Haile, A., et al. 2024. Scaling success: Expanding community-based breeding programs for sheep and goats in Ethiopia. Nairobi, Kenya: ILRI. <https://hdl.handle.net/10568/172417>
- ⁴ Barria, A., Trinh, T.Q., Mahmuddin, M. et al. 2021. A major quantitative trait locus affecting resistance to Tilapia lake virus in farmed Nile tilapia (*Oreochromis niloticus*). *Heredity*, 127: 334–343. <https://doi.org/10.1038/s41437-021-00447-4>
- ⁵ Mrode, R., Ojango, J., Ekine-Dzivenu, C., Aliloo, H., Gibson, J. and Okeyo, M.A. 2021. Genomic prediction of crossbred dairy cattle in Tanzania: A route to productivity gains in smallholder dairy systems. *Journal of Dairy Science*, <https://hdl.handle.net/10568/114617>
- ⁶ S. Bedhiaf-Romdhani, Imen Baazaoui, Emna Arari, Mustapha Lahmar, et al. 2023. Marker-assisted introgression of fecundity mutation into Barbarine sheep breed under low input production system. *Tropical Animal Health and Production*, 55. <https://hdl.handle.net/10568/132882>
- ⁷ E.G. Donkersloot, G.M. Worth, A.F. Yeates, M.D. Littlejohn, et al. 2020. *The benefit of a slick hair coat for heat tolerance in New Zealand dairy cattle*. Newstead, New Zealand: Livestock Improvement Corporation Ltd. <https://www.aaabg.org/aaabghome/AAABG24papers/25Donkersloot24094.pdf>
- ⁸ Jones, C. and Varijakshapanicker, P. 2021. Adapting biofuel technology to convert crop residues into nutritious livestock feed. Innovation Brief. Nairobi, Kenya: ILRI <https://cgspace.cgiar.org/handle/10568/116747>
- ⁹ Jane Wamatu, Muluken Ekule, Kibreab Wodebo, Cherinet Rete, Tesfaye Abiso, Tesfa Belayneh. (18/12/2024). Empowering Youth and Women through Sheep Fattening in Ethiopia. <https://hdl.handle.net/10568/169854>
- ¹⁰ International Center for Tropical Agriculture (CIAT).; International Center for Agricultural Research in the Dry Areas.; International Livestock Research Institute. 2024. Screening, developing, and deploying anti-methanogenic forages into livestock systems in the Global South. 1 p. <https://hdl.handle.net/10568/168748>
- ¹¹ Cardoso Arango, J.A.; Mayorga Cobos, M.J. 2024. Data on impact of soil compaction, drought and their combination on vertical root distribution, morphological traits, and soil water patterns in Urochloa hybrids. <https://doi.org/10.7910/DVN/GDQBQ1> <https://hdl.handle.net/10568/163747>
- ¹² Mainack, M., Folusho, D., Onifade, O. 2024. ILRI News. [New vaccine to protect Mali's sheep and goats against peste des petits ruminants](#). [New vaccine to protect Mali's sheep and goats against peste des petits ruminants](#)
- ¹³ Zannou, O.M., Sow, A.N., Sissoko, B. et al. 2024. Space–Time Analysis of Peste des Petits Ruminants in Mali and Identification of Risk Factors. *Transboundary and Emerging Diseases*, Volume 2024, Article ID 9903861, 10 pages. <https://onlinelibrary.wiley.com/doi/epdf/10.1155/tbed/9903861?msocid=320aec73fdaf6d333e23f8f4fce86c33>
- ¹⁴ Moliso, M.M., Molla, W., Arke, A. 2024. The impact of the herd health interventions in small ruminants in low input production systems in Ethiopia. *Frontiers in Veterinary Science*, 21(11). <https://doi.org/10.3389/fvets.2024.1371571>
- ¹⁵ Ouma, E., Mugabi, D., Ssimbwa, D. et al. 2024. *Pig farmers knowledge enhanced through Pig Smart extension platform in Uganda*. Nairobi, Kenya. International Livestock Research Institute (ILRI). <https://hdl.handle.net/10568/168983>
- ¹⁶ Bullock, R., Dutta Gupta, T., Najjar, D. et al. 2025. Developing approaches for more socially inclusive climate-smart livestock practices. Montpellier: CGIAR. <https://hdl.handle.net/10568/173808>
- ¹⁷ Molina-Botero, I.C., Villegas, D.M., Montoya, A. et al. Effect of a silvopastoral system with *Leucaena diversifolia* on enteric methane emissions, animal performance, and meat fatty acid profile of beef steers. 2024. *Agroforest Systems*, 98(1967–1984). <https://doi.org/10.1007/s10457-024-01046-y>

- ¹⁸ Ali, H. Murshed-E-Jahan Khondker. (2024). Enhancing productivity, nutrition, and climate resilience through integrated aquaculture-agriculture systems. Bayan Lepas, Malaysia: WorldFish (WF). <https://hdl.handle.net/20.500.12348/6226>
- ¹⁹ Obiero, K.O., Ogello, E., Munguti, J. et al. J (2024). Profiling and Prioritizing Climate-Smart Aquaculture Technologies, Innovations, and Management Practices in Kenya. Aquaculture Research. <https://doi.org/10.1016/j.agee.2024.108970>
- ²⁰ Rossignoli, C., Manyise, T., Shikuku, K. et al. 202). Tilapia aquaculture systems in Egypt: Characteristics, sustainability outcomes and entry points for sustainable aquatic food systems. *Aquaculture*, 577. <https://hdl.handle.net/20.500.12348/5595>
- ²¹ Cosecha y manejo de agua por pequeños agricultores ganaderos en Caquetá, Colombia. YouTube (in Spanish). Alliance of Bioversity International and CIAT.
- ²² CGIAR Initiative on One Health. 2024. Trade-offs between wildlife conservation and human health and well-being in Côte d'Ivoire. Montpellier: CGIAR. <https://www.cgiar.org/news-events/news/trade-offs-between-wildlife-conservation-and-human-health-and-well-being-in-cote-divoire/>
- ²³ Grace, D., Bett, B., Cook, E. et al. 2024. Eating wild animals: Rewards, risks and recommendations. Nairobi, Kenya: ILRI. <https://cgspace.cgiar.org/items/1ca9e796-ad7a-4e7d-8bad-5e0a4867ee43>
- ²⁴ Grace, D., Bonfoh, B., Häslér, B., Nguyen-Viet, H. 2024. Editorial: Food safety in low- and middle-income countries. *Frontiers in Sustainable Food Systems*. <https://www.frontiersin.org/research-topics/39835/food-safety-in-low-and-middle-income-countries/magazine#articles>
- ²⁵ Njaramba, J.K., Muloi, D.M., Velde, M.V. et al. 2025. Multi-mycotoxin occurrence and their risk to poultry health in semi-intensive broiler farms in Kenya. *Poultry Science*, 104(5). <https://doi.org/10.1016/j.psj.2025.105008>
- ²⁶ Ambler, K., Cook, Elizabeth A.J., Hoffmann, V. et al. 2024. Impact of a slaughterhouse hygiene intervention in western Kenya. IFPRI Project Note December 2024. Washington, DC: International Food Policy Research Institute. <https://hdl.handle.net/10568/168522>
- ²⁷ Ambler, K., Murphy, M., Dang-Xuan, S. et al. 2024. Business and public health impacts of a food safety rating program among pork vendors in Vietnam. *IFPRI Working Paper December 2024*. Washington, DC: International Food Policy Research Institute. <https://hdl.handle.net/10568/168837>
- ²⁸ Muloi, D.M., Kurui, P., Sharma, G. et al. 2023. Antibiotic quality and use practices amongst dairy farmers and drug retailers in central Kenyan highlands. *Scientific Reports*, 13. <https://doi.org/10.1038/s41598-023-50325-8>
- ²⁹ <https://www.ilri.org/news/kenyas-livestock-experts-propose-revisions-rift-valley-fever-contingency-plan>
- ³⁰ Rao, E.J.O., Mtimet, N., Twine, E. et al. 2018. Farmers' preference for bundled input-output markets and implications for adapted dairy hubs in Tanzania—A choice experiment. *Agribusiness*, 35(3), 358–373. <https://doi.org/10.1002/agr.21565>
- ³¹ Buisson, M.-C., Zane, G., Appiah, S. et al. 2023. Fish cage culture in small water bodies in North East Region of Ghana: Technical and institutional guiding principles for sustainable and inclusive uptake. Report. Colombo, Sri Lanka: International Water Management Institute (IWMI). CGIAR Initiative on Aquatic Foods. <https://cgspace.cgiar.org/items/2f8947ea-e514-4cb0-bc68-a7bd425a745e>
- ³² Dompok, E.B., Rossignoli, C., Griffiths, D. et al. 2023. Impact of adoption of better management practices and nutrition-sensitive training on the productivity, livelihoods and food security of small-scale aquaculture producers in Myanmar. *Food Security*. <https://doi.org/10.1007/s12571-023-01415-y>
- ³³ Sithirith, M., Sao, S., Silva, S.D. et al. 2024. Governance of the food system in the Mekong Delta, Cambodia: Rice, fish, water and NRM. *Technical Report*. Penang, Malaysia: WorldFish. <https://cgspace.cgiar.org/items/1bf75411-241b-4852-94bd-07b307e0f468>
- ³⁴ Ramachandran, M., Saito, O., Lam, R.D. et al. 2023. Why the data gaps in Indian aquaculture need to be addressed to contribute to its efficient and sustainable development. *Report*. Bayan Lepas, Malaysia: WorldFish (WF). <https://hdl.handle.net/10568/137679>
- ³⁵ Bullock, R., DuttaGupta, T. 2024. Community conversations to foster social inclusion in climate adaptation: IPSR Innovation Profile. *Report*. First edition, September 2024. Montpellier: CGIAR System Organization. <https://hdl.handle.net/10568/162674>
- ³⁶ Bullock, R., Dutta Gupta, T., Najjar, D. et al. 2025. Developing approaches for more socially inclusive climate-smart livestock practices. *Report*. Montpellier: CGIAR. <https://doi.org/10.3389/fpubh.2024.1345273>

-
- ³⁷ Nyakundi, E. 2025. Early warning early action and WCIS for pastoralists. Report. Presented at Weather and Climate Information Services for Pastoralists – Evidence Review and Distillation Workshop, Nairobi, 18–19 March 2025. Nairobi: Save the Children. <https://hdl.handle.net/10568/174174>
- ³⁸ Vyas, S., Burra, D.D., Calvin, K. et al. 2024. Impact of Shamba Shape Up and iShamba on livestock and climate resilience. *Technical Report*. CGIAR Initiative on Livestock and Climate. 21 p. <https://hdl.handle.net/10568/159559>
- ³⁹ Tilley, A., Lam, R.D., Lozano, D. et al. 2024. The impacts of digital transformation on fisheries policy and sustainability: Lessons from Timor-Leste. *Environmental Science and Policy*, 153. <https://doi.org/10.1016/j.envsci.2024.103536> <https://hdl.handle.net/20.500.12348/5839>
- ⁴⁰ Tilley, A., Dos Reis Lopes, J., Wilkinson, S. et al. 2020. PeskaAS: A near-real-time, open-source monitoring and analytics system for small-scale fisheries. *PLoS ONE*, 15(11): e0234760. <https://hdl.handle.net/20.500.12348/4317>
- ⁴¹ Bonilla-Cedrez, C., Steward, P., Rosenstock, T.S. et al. 2023. Priority areas for investment in more sustainable and climate-resilient livestock systems. *Nature Sustainability*, 6, 1279–1286. <https://doi.org/10.1038/s41893-023-01173-https://doi.org/10.1038/s41893-023-01161-1>
- ⁴² Rossignoli, C., Manyise, T., Shikuku, K. et al. 2023. Tilapia aquaculture systems in Egypt: Characteristics, sustainability outcomes and entry points for sustainable aquatic food systems. *Aquaculture*, 577. <https://doi.org/10.1016/j.aquaculture.2023.739956>
- ⁴³ Marshall, K., Poole, J., Njehu, A. and Baltenweck, I. 2022. Monitoring, evaluation and learning within the CGIAR Initiative Sustainable Animal Productivity for Livelihoods, Nutrition and Gender Inclusion (SAPLING). ILRI Research Brief 117. Nairobi, Kenya: ILRI. <https://hdl.handle.net/10568/128150>
- ⁴⁴ Ouma, E., Marshall, K., Johnson, N. et al. 2023. Theory of change for the pig value chain in Uganda, developed for the CGIAR Initiative Sustainable Animal Productivity for Livelihoods, Nutrition and Gender Inclusion. *Report*. Nairobi, Kenya: ILRI. <https://hdl.handle.net/10568/131880>
- ⁴⁵ Ouma, E., Marshall, K., Lukuyu, B. et al. 2023. SAPLING Theory of Change reflection workshop report on dairy and pig value chains in Uganda. *Report*. Nairobi, Kenya: International Livestock Research Institute (ILRI). <https://hdl.handle.net/10568/138121>
- ⁴⁶ Businge, M., Oba, P., Ahumuza, R. et al. 2024. Lessons and early outcomes: Report from SAPLING Theory of Change reflection workshop held in Kampala, Uganda. *Report*. Nairobi, Kenya: ILRI. <https://hdl.handle.net/10568/170140>